Test Series No.

1.11

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 17/03/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A particle moves in a straight line with a constant acceleration. It changes its velocity from 10 m/s to 20 m/s while passing through a distance of 135 m in t seconds. The value of t is:

(1) 9 (2) 10
(3) 1.8 (4) 12

2. Two long straight parallel conductors separated by a distance d carry currents I_1 and I_2 in the same direction. They exert a force F on each other. If the current in one is doubled and the distance between them is tripled, the new force between them will be:

(1) $\frac{F}{6}$ (2) $\frac{2F}{3}$
(3) $\frac{3F}{2}$ (4) $\frac{F}{3}$

3. Given below are two statements:

Statement I: In an adiabatic expansion of an ideal gas, the product of absolute pressure and volume decreases.

Statement II: In an adiabatic expansion, the internal energy of the gas decreases.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

4. An electromagnetic wave of frequency 3 MHz passes from vacuum into a dielectric medium with relative permittivity $\epsilon_r = 4$. The change in wavelength is: (consider nonmagnetic medium)

(1) 50 m (2) 100 m
(3) 25 m (4) 0 m

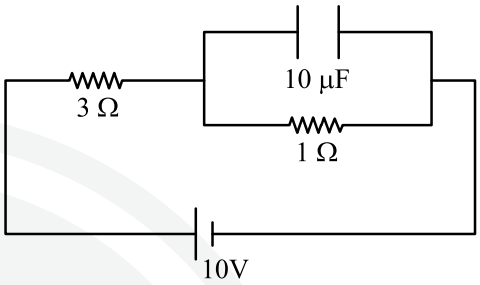
5. In a Young's double slit experiment, the separation between the slits is halved and the distance between the slits and the screen is doubled. The fringe width becomes:

(1) Halved
(2) Quadrupled
(3) Doubled
(4) Unchanged

6. A body of mass 60 g experiences a gravitational force of 3.0 N when placed at a particular point. The magnitude of the gravitational field intensity at that point is:

(1) 0.05 N/kg (2) 20 N/kg
(3) 180 N/kg (4) 50 N/kg

7. In the given circuit, what is the charge stored on the capacitor in steady state?



(1) 2.5 μC (2) 5 μC
(3) 10 μC (4) 25 μC

8. A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is:

(1) $11RT$ (2) $15RT$
(3) $9RT$ (4) $4RT$

9. Two resistors of resistance $R_1 = (4 \pm 0.8)\Omega$ and $R_2 = (4 \pm 0.4)\Omega$ are connected in parallel. The equivalent resistance of the parallel combination is:

(1) $(2 \pm 0.3)\Omega$
(2) $(2 \pm 0.4)\Omega$
(3) $(2 \pm 0.1)\Omega$
(4) $(2 \pm 0.6)\Omega$

10. The magnetic flux linked with a coil (in Weber) is given by the equation $\phi = 5t^2 + 3t + 16$. The magnitude of the average induced emf in the coil between $t = 2$ s and $t = 6$ s will be:

(1) 33 V
(2) 27 V
(3) 10 V
(4) 43 V

[Contd...

11. An electron moves in a circular orbit with a uniform speed v. It produces a magnetic field B at the center of the circle. The radius of the circle is proportional to:

(1) $\sqrt{\frac{v}{B}}$ (2) $\frac{v}{B}$

(3) $\frac{B}{v}$ (4) $\sqrt{\frac{B}{v}}$

12. For a given nuclear reaction, the particle X is:

$_6C^{11} \rightarrow {}_5B^{11} + \beta^+ + X$

(1) Neutron
(2) Electron
(3) Proton
(4) None of these

13. A body of mass m is taken from the surface of the earth to a height equal to the radius of the earth (R). The change in potential energy is:

(1) mgR (2) $\frac{1}{2}mgR$

(3) $2mgR$ (4) $\frac{1}{4}mgR$

14. A uniform rope of length L and mass M is placed on a smooth horizontal table. A horizontal force F is applied at one end. The tension in the rope at a distance x from the applied end is:

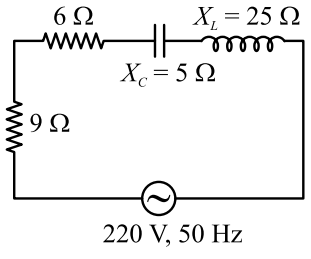
(1) $F\left(1-\frac{x}{L}\right)$

(2) $F\left(\frac{x}{L}\right)$

(3) $F\left(1+\frac{x}{L}\right)$

(4) F

15. Power factor of LCR series circuit shown in the figure is:



(1) 0.2 (2) 0.8

(3) 0.4 (4) 0.6

16. Given below are two statements:

Statement I: A convex mirror always produces a virtual, erect, and diminished image for a real object.

Statement II: A concave mirror always produces a real, inverted, and magnified image for a real object.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

17. Match the physical quantities given in **List-I** with their correct dimensional formulae in **List-II**.

	List-I		**List-II**
(A)	Electrical Capacitance	(I)	$[M^1L^2T^{-1}]$
(B)	Magnetic Permeability (μ_0)	(II)	$[M^{-1}L^{-2}T^4A^2]$
(C)	Planck's Constant (h)	(III)	$[M^1L^{-1}T^{-2}]$
(D)	Young's Modulus (Y)	(IV)	$[M^1L^1T^{-2}A^{-2}]$

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-I, C-III, D-II

18. Two polaroids are placed in the path of unpolarized beam of intensity I_0. The axes of the two polaroids are initially parallel to each other. If the second polaroid is rotated by an angle 60°, the intensity of emerging light will be:

(1) $\frac{I_0}{8}$ (2) $\frac{I_0}{4}$

(3) $\frac{I_0}{2}$ (4) $\frac{3I_0}{8}$

19. A car of mass 1000 kg negotiates a banked curve of radius 90 m on a frictionless road. If the banking angle is 45°, the speed of the car is: ($g = 10\ m/s^2$)

(1) 20 m/s
(2) 30 m/s
(3) 5 m/s
(4) 10 m/s

[Contd...

20. A metallic wire of length 2 m and cross-sectional area $4\ mm^2$ is stretched by an external force. If the elastic potential energy stored in the wire is 0.4 J when the elongation produced is 2 mm, then the Young's modulus of the material of the wire is:

(1) $2.0\times10^{11}\ N/m^2$

(2) $1.0\times10^{11}\ N/m^2$

(3) $4.0\times10^{10}\ N/m^2$

(4) $5.0\times10^{10}\ N/m^2$

21. An open organ pipe has a fundamental frequency of 300 Hz. The first overtone of a closed organ pipe has the same frequency as the first overtone of this open pipe. The length of the closed pipe is roughly: (Speed of sound $v = 330$ m/s)

(1) 41 cm (2) 82 cm

(3) 20 cm (4) 15 cm

22. A proton and an alpha particle enter a uniform magnetic field perpendicularly with the same momentum. The ratio of the radii of their circular paths $r_p : r_\alpha$ is:

(1) 1:2 (2) 2:1

(3) 1:1 (4) 4:1

23. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The average velocity of a particle in SHM in one complete cycle is zero.

Reason (R): Average speed of a particle in SHM in one complete cycle is zero.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

24. In a thermodynamic process, the pressure of a fixed mass of gas is changed in such a manner that the gas releases 20 J of heat and 8 J of work is done on the gas. If the initial internal energy of the gas was 30 J, the final internal energy will be:

(1) 18 J (2) 42 J

(3) 58 J (4) 2 J

25. Two point charges $+9e$ and $+e$ are kept 16 cm apart. At what distance from the charge $+9e$ should a third charge q be placed so that it remains in equilibrium? (e is the charge on electron)

(1) 12 cm (2) 8 cm

(3) 4 cm (4) 10 cm

26. A planet moves around the sun in an elliptical orbit. When it is at the perihelion (closest point), its distance is r_1 and velocity is v_1. When it is at the aphelion (farthest point), its distance is r_2. Its velocity at aphelion is:

(1) $v_1\left(\frac{r_1}{r_2}\right)$

(2) $v_1\left(\frac{r_2}{r_1}\right)$

(3) $v_1\left(\frac{r_1}{r_2}\right)^2$

(4) v_1

27. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is: ($g = 10\ m/s^2$)

(1) 340 m (2) 320 m

(3) 300 m (4) 360 m

28. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion: The resistance of an intrinsic semiconductor decreases with an increase in temperature.

Reason: With an increase in temperature, the relaxation time of charge carriers increases significantly in semiconductors.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

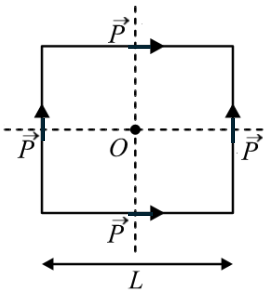
(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

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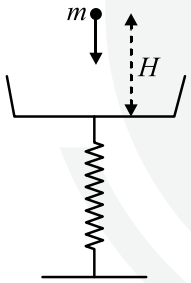
29. Four short electric dipole each having dipole moment $\vec{P}$ are placed at the centre of edge of a square of side L as shown. What is the magnitude of net electric field due to all the dipoles at the centre of the square?



(1) $\frac{8kP}{L^3}$ (2) $\frac{8\sqrt{2}kP}{L^3}$

(3) $\frac{16\sqrt{2}kP}{L^3}$ (4) Zero

30. Particles each of mass m are falling into massless weighing pan from height H at a rate of n particles per second. Find reading of weighing pan at time t seconds, if collision of particle with weighing pan is perfectly inelastic. (Neglect oscillation of weighing pan and g is acceleration due to gravity)



(1) $mgHt + mgnt$ (2) $mn(gt + \sqrt{2gH})$

(3) $mng\sqrt{2gH}$ (4) None of these

31. The frequency of the first line of the Lyman series of the hydrogen atom is ν. The frequency of the first line corresponding to the singly ionised helium atom is

(1) 8 ν (2) 4 ν

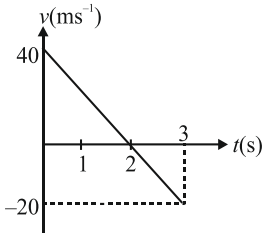
(3) 2 ν (4) ν

32. 32 cells, each of emf 3 V, are connected in series and kept in a box. Due to some cells are wrongly connected, the combination externally shows an emf of 84 V. The number of cells reversed in the combination is

(1) 0 (2) 2

(3) 4 (4) 8

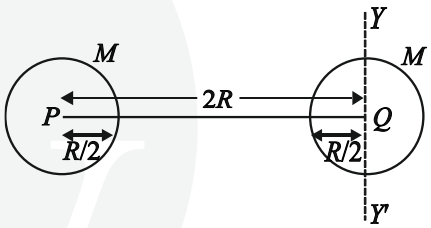
33. Velocity-time graph of a particle of mass 3 kg moving in a straight line is shown in the figure. The work done by all the forces acting on the particle in 3s is:



(1) –1800 J (2) +1800 J

(3) +3200 J (4) –3200 J

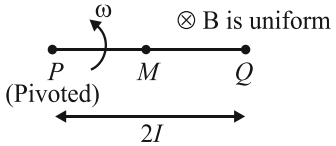
34. Two solid spheres each of mass M and radius $\frac{R}{2}$ are connected with a massless rod PQ of length $2R$ with the centers of the spheres connected with the ends of the rod as shown in figure. The moment of inertia of the system about an axis YY' passing through the centre of one of the spheres and perpendicular to the rod is:



(1) $\frac{21}{5}MR^2$ (2) $\frac{2}{5}MR^2$

(3) $\frac{5}{2}MR^2$ (4) $\frac{5}{21}MR^2$

35. A uniform rod PQ rotates about pivoted point P in uniform magnetic field B as shown. The emf induced between M and Q, if the potential between P and Q is 100 V, will be: (M is mid point of P and Q)



(1) 25 V (2) 50 V

(3) 75 V (4) 100 V

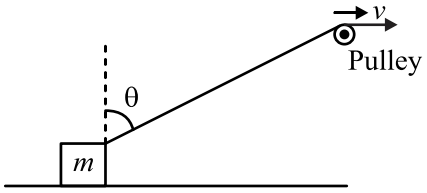
36. 10 gm of ice cubes at 0°C are released in a tumbler (water equivalent 55 g) at 40°C. Assuming that negligible heat is taken from the surroundings, the approximate temperature of water in the tumbler becomes nearly (L = 80 cal/g)

(1) 31°C (2) 22°C

(3) 19°C (4) 15°C

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37. If a block is dragged on a smooth plane with the help of thin rope which moves with velocity v as shown in the figure, then horizontal velocity of block at given instant is:



(1) v (2) $v\sin\theta$

(3) $\frac{v}{\sin\theta}$ (4) $\frac{v}{\cos\theta}$

38. The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m, is: (where k is Boltzmann constant)

(1) $\frac{h}{\sqrt{3mkT}}$ (2) $\frac{2h}{\sqrt{3mkT}}$

(3) $\frac{h}{\sqrt{mkT}}$ (4) $\frac{h}{\sqrt{2mkT}}$

39. Consider an intrinsic semiconductor in which density of conduction electrons is 7.07×10^{15} m^{-3}. If it is doped with indium then density of holes becomes 5×10^{22} m^{-3}, then density of conduction electrons in doped semiconductor is nearly:

(1) 10^5 per m^3 (2) 10^9 per m^3

(3) 10^7 per m^3 (4) 10^3 per m^3

40. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and the eye-piece is 36 cm and the final image is formed at infinity. The focal length f_0 of the objective and f_e, of the eye-piece are:

(1) $f_0 = 45$ cm and $f_e = 10$ cm

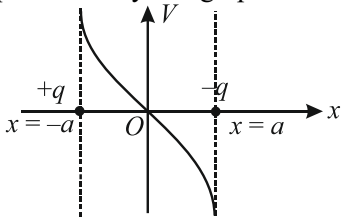
(2) $f_0 = 50$ cm and $f_e = 10$ cm

(3) $f_0 = 7.2$ cm and $f_e = 5$ cm

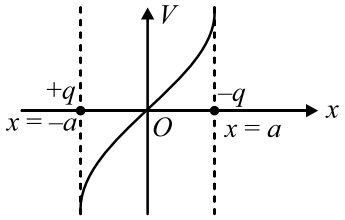
(4) $f_0 = 30$ cm and $f_e = 6$ cm

41. A positive charge q and a negative charge $-q$ are placed at $x = -a$ and $x = +a$ respectively. The variation of electrostatic potential (V) along x-axis is represented by the graph:

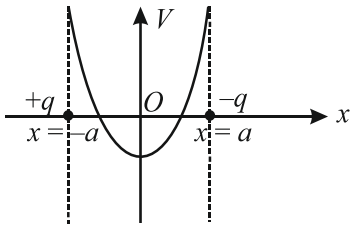
(1)



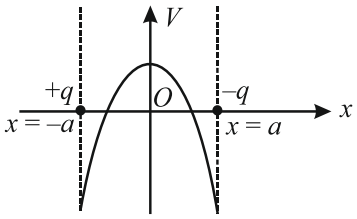
(2)



(3)



(4)



42. The instantaneous angular position (θ) of a point on a rotating wheel is given by the equation $\theta(t) = (2t^3 - 6t^2)$ radian, (where t is time in second). The torque on the wheel becomes zero at:

(1) $t = 1$ s (2) $t = 0.5$ s

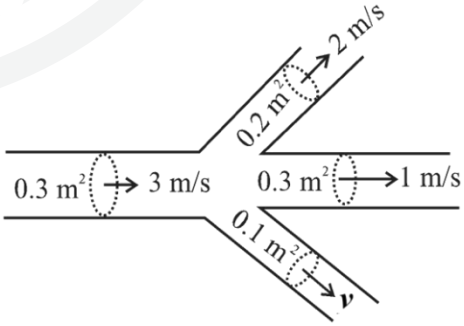
(3) $t = 0.25$ s (4) $t = 2$ s

43. A surface has area vector $\vec{A} = (2\hat{i} + 3\hat{j})$ m^2. The flux of an electric field through it, if the electric field is $\vec{E} = 4\hat{i}\,\frac{\text{V}}{\text{m}}$, is:

(1) 8 V-m (2) 12 V-m

(3) 20 V-m (4) Zero

44. An incompressible liquid flows through channels of pipe as shown in figure. Speed 'v' of the fluid from pipe of cross section area 0.1 m^2 is;



(1) 4 m/sec (2) 3 m/sec

(3) 1 m/sec (4) 2 m/sec

45. Two small bar magnets are placed in a line (end-to-end) at certain distance d apart. If the length of each magnet is negligible compared to d, the force between them will be inversely proportional to:

(1) d^2 (2) d

(3) d^3 (4) d^4

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46. The electronegativity of aluminium is most similar to:
(1) Carbon
(2) Beryllium
(3) Boron
(4) Lithium

47. The functional group that shows negative resonance effect (–R) is:
(1) NH_2 (2) –OH
(3) –COOH (4) –OR

48. For the following reactions, equilibrium constants are given:

$S(s) + O_2(g) \rightleftharpoons SO_2(g)$; $K_1 = 10^{40}$

$2S(s) + 3O_2(g) \rightleftharpoons 2SO_3(g)$; $K_2 = 10^{100}$

The equilibrium constant for the reaction

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is:

(1) 10^{20} (2) 10^{60}
(3) 10^{100} (4) 10^{-20}

49. Identify the process in which change in oxidation state per atom is five:
A. $Cl_2 \rightarrow ClO_3^-$
B. $MnO_4^- \rightarrow Mn^{2+}$
C. $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$
D. $C_2O_4^{2-} \rightarrow 2CO_2$
(1) C only
(2) B and C only
(3) A and B only
(4) B and D only

50. Potassium ferrocyanide solution gives a Prussian blue colour when added to:
(1) $FeSO_4$ (2) $FeCl_3$
(3) $CoCl_2$ (4) $ZnSO_4$

51. The molarity of an aqueous solution containing 2.925 g of NaCl in 250 mL solution is:
(Given: Na = 23, Cl = 35.5 g mol^{-1})
(1) 0.02 M (2) 0.20 M
(3) 2.0 M (4) 4.0 M

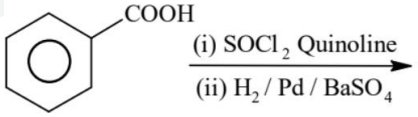
52. The osmotic pressure of a dilute solution of ionic compound XY in water is three times that of 0.02 M $BaCl_2$. Assuming complete dissociation, the concentration of XY is:
(1) 0.03 M
(2) 0.06 M
(3) 0.09 M
(4) 0.18 M

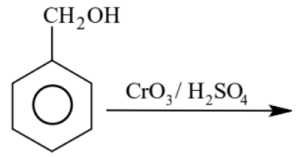
53. A first order reaction has rate constant $k = 2.31 \times 10^{-3}\ s^{-1}$.
The number of **correct** statements is:
(A) Half-life is 300 s.
(B) Time for 25% remaining is 600 s.
(C) Time for 12.5% remaining is 900 s.
(D) Unit of k is s^{-1}.
(E) Rate and k have same unit.
(1) 2 (2) 3
(3) 4 (4) 5

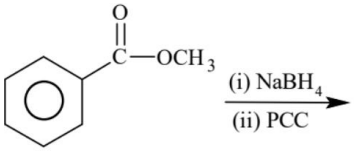
54. The nature of oxide (TeO_2) and hydride (H_2Te) formed by Tellennium, respectively, are:
(1) Oxidising and acidic
(2) Oxidising and basic
(3) Reducing and acidic
(4) Reducing and basic

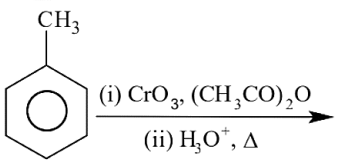
55. The set of **correct** statements is:
(i) Manganese exhibits +7 oxidation state in permanganate.
(ii) Ruthenium and Osmium exhibit +8 oxidation state in their oxides.
(iii) Sc shows +4 oxidation state which is oxidising in nature.
(iv) Chromium shows oxidising nature in +6 oxidation state.
(1) (i) and (ii) only
(2) (i), (ii) and (iv) only
(3) (ii), (iii) and (iv) only
(4) (i), (iii) and (iv) only

56. Which of the following reactions will **not** yield benzaldehyde as a product?

(A) 

(B) 

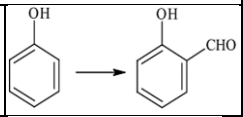
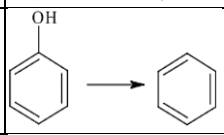
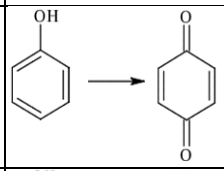
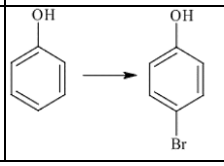
(C) 

(D) 

(1) (B) and (C) (2) (C) and (D)
(3) (A) and (D) (4) (A) and (C)

[Contd...

57. Match **List-I** with **List-II**.

List-I		List-II	
(A)		(I)	Br_2 in CS_2
(B)		(II)	$Na_2Cr_2O_7/H_2SO_4$
(C)		(III)	Zn
(D)		(IV)	$CHCl_3/NaOH$

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-III, C-I, D-IV
(4) A-IV, B-II, C-III, D-I

58. 2,4-DNP test is used to identify:

(1) Halides (2) Amines
(3) Ketones (4) Ethers

59. The increasing order of acidic nature of amino acids in aqueous solution is:

Gly, Asp, Lys, Arg

(1) Asp < Gly < Lys < Arg
(2) Gly < Asp < Lys < Arg
(3) Asp < Gly < Arg < Lys
(4) Arg < Lys < Gly < Asp

60. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;

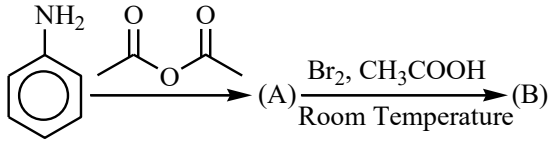
Assertion (A): Nitration of aniline gives a noticeable amount of m-nitroaniline along with o- and p-nitroaniline.

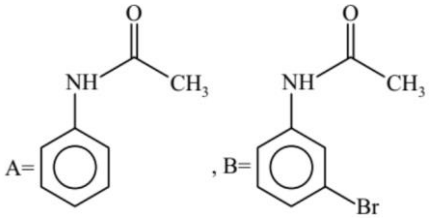
Reason (R): The nitrating mixture is strongly acidic and converts aniline into anilinium ion, which is meta-directing.

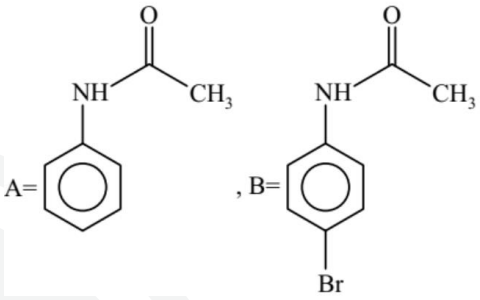
In the light of the above statements, choose the **correct** answer from the options given below:

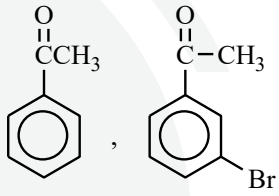
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

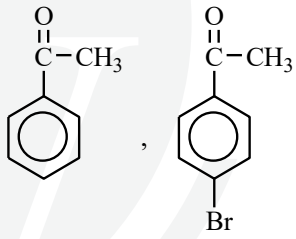
61. The major products (A) and (B) formed in the following reaction sequence are:



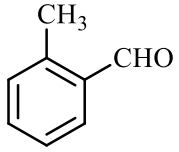
(1) 

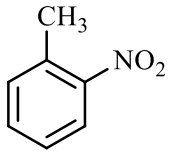
(2) 

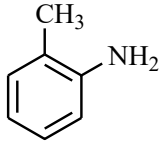
(3) 

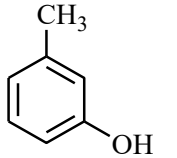
(4) 

62. Halogenation of which one of the following will yield m-substituted product with respect to methyl group as a major product?

(1) 

(2) 

(3) 

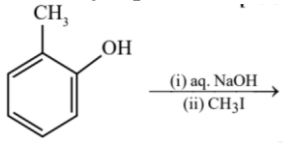
(4) 

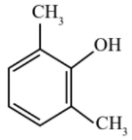
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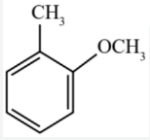
63. Homoleptic octahedral complexes of a metal ion M^{3+} with monodentate ligands L_1, L_2, L_3 absorb orange, green, violet light respectively. The increasing order of ligand strength is:
(1) $L_3 < L_2 < L_1$ (2) $L_1 < L_2 < L_3$
(3) $L_2 < L_1 < L_3$ (4) $L_1 < L_3 < L_2$

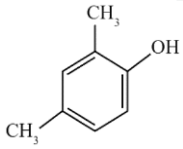
64. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Size of Cm^{3+} ion is less than U^{3+} ion.
Reason R: This is a consequence of actinoid contraction.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true but R is not the correct explanation of A.
(2) A is true but R is false.
(3) Both A and R are true and R is the correct explanation of A.
(4) A is false but R is true.

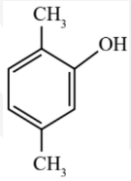
65. The major product of the following reaction is:



(1) 

(2) 

(3) 

(4) 

66. Potassium permanganate on heating at about 520 K gives a manganate product which is:
(1) Paramagnetic and colourless
(2) Diamagnetic and colourless
(3) Diamagnetic and green
(4) Paramagnetic and green

67. A reagent that gives a brilliant red precipitate with Ni^{2+} ions in ammoniacal/basic medium is:
(1) Sodium nitroprusside
(2) Neutral $FeCl_3$
(3) meta-Dinitrobenzene
(4) Dimethyl glyoxime

68. Which of the following is a water-soluble vitamin that is **not** excreted easily?
(1) Vitamin B_1 (2) Vitamin B_2
(3) Vitamin B_6 (4) Vitamin B_{12}

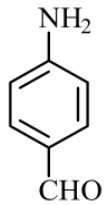
69. Identify the **incorrect** pair from the following:
(1) Fluorspar — CaF_2
(2) Cryolite — Na_2AlF_6
(3) Fluorapatite — $Ca_5(PO_4)_3F$
(4) Carnallite — $KCl.MgCl_2.6H_2O$

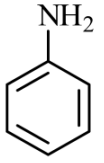
70. The statement that is **INCORRECT** about interstitial compounds is:
(1) They have metallic conductivity
(2) They have high melting points
(3) They are very hard
(4) They are poor electrical conductors

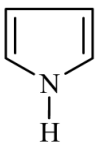
71. Given below are two statements:
Statement-I: In transcription, RNA polymerase synthesizes RNA in the 3' → 5' direction on the template strand.
Statement-II: In transcription, the template strand is read in the 3' → 5' direction.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement-I and Statement-II are correct
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I correct, Statement-II incorrect
(4) Statement-II correct, Statement-I incorrect

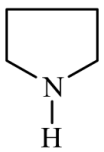
72. A nitrogen-containing aromatic compound A is reduced with Sn/HCl to give an aromatic amine, which on treatment with cold HNO_2 (0–5°C) forms an unstable intermediate B. Compound B couples with phenol in alkaline medium to give a brightly coloured azo dye C having molecular formula $C_{12}H_{10}N_2O$. The structure of A is:
(1) Nitrobenzene
(2) p-Nitroanisole
(3) p-Nitrotoluene
(4) p-Nitrophenol

73. Strongest base among the following is:

(1) 

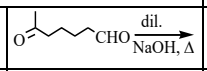
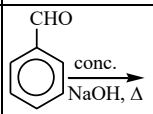
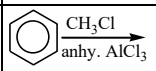
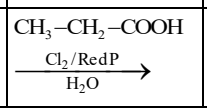
(2) 

(3) 

(4) 

[Contd...

74. Match **List-I** with **List-II**.

List-I		List-II	
(A)		(I)	HVZ reaction
(B)		(II)	Friedel Craft's alkylation
(C)		(III)	Cannizzaro reaction
(D)		(IV)	Intramolecular aldol condensation

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-II, B-III, C-IV, D-I

75. During the preparation of Mohr's salt solution ($(NH_4)_2Fe(SO_4)_2.6H_2O$), which acid is added to prevent hydrolysis of Fe^{2+} ion?
(1) Concentrated H_2SO_4
(2) Dilute HNO_3
(3) Dilute H_2SO_4
(4) Dilute HCl

76. Number of moles of methane required to produce 8.8 g CO_2 on complete combustion is $x \times 10^{-2}$ moles. The value of x is:
(1) 10 (2) 20
(3) 40 (4) 5

77. According to Bohr's atomic theory for hydrogen-like species, which statements are **correct**?
(A) Kinetic energy of electron $\propto \frac{Z^2}{n^2}$
(B) The product of velocity and principal quantum number $\propto Z$
(C) Frequency of revolution $\propto \frac{Z^2}{n^3}$
Choose the *most appropriate* answer:
(1) A and B only (2) A only
(3) A, C and B (4) C only

78. The extra stability of a half-filled subshell is due to:
(A) Symmetrical distribution of electrons
(B) Smaller electron–electron repulsion due to less pairing
(C) Presence of same-spin electrons in non-degenerate orbitals
(D) Larger exchange energy
(E) Greater shielding of electrons by one another
Identify the correct statements:
(1) (B), (D) and (E) only
(2) (A), (B), (D) and (E) only
(3) (A), (B) and (D) only
(4) (A), (C) and (D) only

79. Among the following, the number of state variables (state functions) is:
Internal energy (U), Heat (q), Volume (V), Enthalpy (H)
(1) 1 (2) 2
(3) 3 (4) 4

80. Which one of the following molecules has maximum dipole moment?
(1) NF_3 (2) H_2O
(3) CO_2 (4) BF_3

81. Match **List-I** with **List-II**.

List-I (Methyl halide)		List-II (Dipole moment)	
(A)	CH_3F	(I)	1.860 D
(B)	CH_3Cl	(II)	1.636 D
(C)	CH_3Br	(III)	1.847 D
(D)	CH_3I	(IV)	1.830 D

Choose the **correct** answer from the options given below:
(1) A-I; B-II; C-IV; D-III
(2) A-II; B-IV; C-III; D-I
(3) A-III; B-II; C-I; D-IV
(4) A-III; B-I; C-IV; D-II

82. ΔS_{surr} for the oxidation of iron, $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ at 298 K is:
[Given: $\Delta_r H^o$ for the reaction is -1648×10^3 J mol^{-1})
(1) 549 JK^{-1} mol^{-1} (2) 5530 JK^{-1} mol^{-1}
(3) 4980 JK^{-1} mol^{-1} (4) 2348 JK^{-1} mol^{-1}

83. Match **List-I** with **List-II**.

List-I (Battery)		List-II (Cathode reaction)	
(A)	Mercury cell	(I)	$PbO_2 \rightarrow PbSO_4$
(B)	Lead storage battery during discharging	(II)	$O_2 \rightarrow OH^-$
(C)	Dry cell	(III)	$HgO \rightarrow Hg$
(D)	Fuel cell	(IV)	$MnO_2 \rightarrow MnO(OH)$

Choose the **correct** answer from the options given below:
(1) A-II; B-IV; C-I; D-III
(2) A-II; B-I; C-III; D-IV
(3) A-III; B-I; C-II; D-IV
(4) A-III; B-I; C-IV; D-II

[Contd...

84. For the group detection test of basic radicals using appropriate group reagents, which of the following ions can be precipitated as carbonates?

A. Ba^{2+}

B. Zn^{2+}

C. Pb^{2+}

The **correct** option(s) is/are;

(1) A only

(2) A and B only

(3) A, B and C

(4) B and C only

85. Among the following statements;

A. s-block and p-block elements are collectively known as representative elements.

B. Group-16 elements are known as chalcogens.

C. Atomic radius of noble gases is expressed as van der Waal's radius.

The **correct** statement(s) is/are;

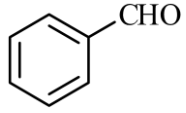
(1) A and B only

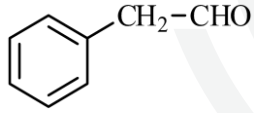
(2) A and C only

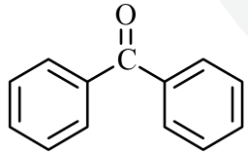
(3) B and C only

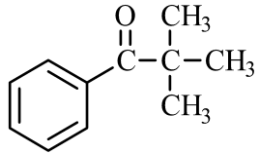
(4) A, B and C

86. Tautomerism is shown by:

(1) 

(2) 

(3) 

(4) 

87. The **correct** relation between cryoscopic constant and freezing point is:

(1) $K_f = \frac{R \times M_1 \times T_f^2}{1000 \times \Delta_{fus}H}$

(2) $K_f = \frac{R \times M_1 \times T_f}{1000 \times \Delta_{fus}H}$

(3) $K_f = \frac{R \times M_1 \times \Delta_{fus}H}{1000 \times T_f^2}$

(4) $K_f = \frac{R \times M_1 \times \Delta_{fus}H}{1000 \times T_f}$

88. The standard electrode potential of Fe^{2+}/Fe and Fe^{3+}/Fe^{2+} are given below:

$Fe^{2+} + 2e^- \rightarrow Fe;\ E^o = -0.44\ V$

$Fe^{3+} + e^- \rightarrow Fe^{2+};\ E^o = 0.77\ V$

The value of $E^o_{Fe^{3+}/Fe}$ is:

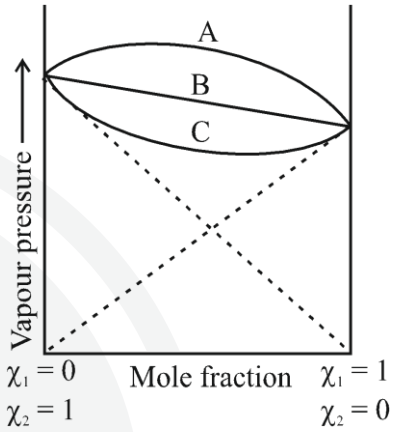
(1) 0.11 V

(2) 0.03 V

(3) –0.11 V

(4) –0.03 V

89. In the given graph, A, B and C corresponds to:



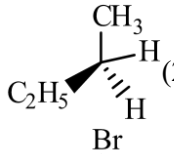
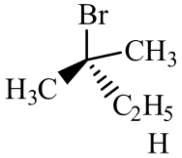
(1) A = Ideal solution
B = Solution showing positive deviation from Raolult's law
C = Solution showing negative deviation from Raoult's law

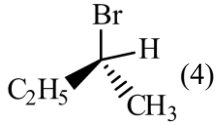
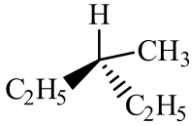
(2) A = Solution showing positive deviation from Raolult's law
B = Ideal solution
C = Solution showing negative deviation from Raoult's law

(3) A = Ideal solution
B = Solution showing negative deviation from Raoult's law
C = Solution showing positive deviation from Raolult's law

(4) A = Solution showing negative deviation from Raoult's law
B = Ideal solution
C = Solution showing positive deviation from Raolult's law

90. Chiral molecule among the following is:

(1)  (2) 

(3)  (4) 

[Contd...

91. Identify the **true** statement for the quiescent stage of the cell cycle.
(1) Cells in this stage are metabolically inactive.
(2) Cells enter this stage after exiting the G_2 phase.
(3) Cells in this stage no longer proliferate unless called on to do so.
(4) This stage is characterized by DNA synthesis.

92. Given below are two statements:
Statement I: DNA contains ribose sugar.
Statement II: In a nucleotide, phosphate group is attached at 5′ position of the sugar.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

93. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The population interaction between a sparrow feeding on seeds is predation.
Reason (R): In an ecosystem, a sparrow can be primary as well as secondary consumer.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

94. Choose the **incorrect** statement about pericycle.
(1) In a monocot root, all tissues inside it are considered together as stele.
(2) Initiation of lateral roots and vascular cambium in dicot root takes place in its cells.
(3) In dicot stem, it is in the form of semi-lunar patches of sclerenchyma.
(4) It is a part of the ground tissue system.

95. An organic farmer observes that his brassica and fruit tree crops are being damaged by butterfly insect larvae. To manage this pest problem without using chemical pesticides, which of the following microbial biocontrol agents would be most effective and specific?
(1) *Trichoderma*
(2) *Bacillus thuringiensis*
(3) *Nucleopolyhedrovirus*
(4) Ladybird beetle

96. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Red alga	(I)	*Polysiphonia*
(B)	Green alga	(II)	*Chara*
(C)	Brown alga	(III)	*Sargassum*
(D)	Blue-green alga	(IV)	*Nostoc*

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-IV, C-II, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-I, C-II, D-IV
(4) A-I, B-II, C-III, D-IV

97. All the following are the suspecies of tiger that have recently become extinct; **except**:
(1) Javan.
(2) Caspian.
(3) Dodo.
(4) Bali.

98. Axile placentation is present in;
A. Pea
B. China rose
C. Mustard
D. *Primrose*
E. Lemon
Choose the **correct** option.
(1) B and E
(2) A and C
(3) C and D
(4) B and C

99. An aquatic plant which does **not** use water as pollinating agent, is;
(1) *Hydrilla*.
(2) *Vallisneria*.
(3) *Eicchornia*.
(4) *Zostera*.

100. Arrange the following intermediates of glycolysis in the correct sequence of their appearance in the pathway.

A. Fructose-6-phosphate
B. 3-phosphoglyceric acid
C. Fructose 1, 6-bisphosphate
D. Phosphoenolpyruvate
E. Glucose-6-phosphate

Choose the **correct** option.

(1) C → A → E → B → D
(2) A → E → C → B → D
(3) E → C → A → D → B
(4) E → A → C → B → D

101. In the Mendelian disorder shown below, the cause of disease is;




(1) mutation in the gene that codes for the enzyme phenyl alanine hydroxylase.
(2) substitution of glutamate by valine at the sixth position of the beta globin chain of the haemoglobin molecule.
(3) mutation or deletion of HBA1 and HBA2 genes on chromosome 16.
(4) mutation or deletion of HBB gene on chromosome 11.

102. Identify the **false** statement about fig-wasp mutualism.

(1) The wasp lays eggs in fig flower.
(2) These both have a tight one-to-one relationship.
(3) The wasp pollinates the fig inflorescence.
(4) A given fig species can be pollinated only by its 'partner' wasp species and no other species.

103. The TCA cycle occurs in the;

(1) lumen of thylakoid.
(2) stroma of chloroplast.
(3) matrix of mitochondria.
(4) cell cytoplasm.

104. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Mycoplasma are organisms that can survive without oxygen.

Reason (R): Mycoplasma are the smallest living cells known and completely lack a cell wall.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

105. 'A' is primarily responsible for apical dominance, which can be overcome by 'B'. A and B, respectively, are;

(1) Gibberellin and Abscisic acid
(2) Cytokinin and Ethylene
(3) Abscisic acid and Ethylene
(4) Auxin and Cytokinin

106. Katherine Esau reported that the _____ spreads through a plant via the phloem.

(1) tobacco mosaic virus
(2) curly top virus
(3) QB bacteriophage
(4) adenovirus

107. Given below are two statements:

Statement I: In C_4 plants, C_3 pathway is the main biosynthetic pathway.

Statement II: In C_3 plants, photorespiration does not occur.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

108. Match **List-I** with **List-II** for housefly.

List-I		List-II	
(A)	Family	(I)	Insecta
(B)	Order	(II)	Arthropoda
(C)	Class	(III)	Muscidae
(D)	Phylum	(IV)	Diptera

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-II, C-III, D-IV

109. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Lactose is the inducer for the *lac* operon.

Reason (R): The repressor of the *lac* operon is inactivated by interaction with lactose, which allows RNA polymerase access to the promoter and transcription proceeds.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

110. A faulty ribosome or its subunit that will compromise protein synthesis in chloroplast, is;

(1) 80S
(2) 60S
(3) 40S
(4) 30S

111. Regarding the process of decomposition, identify the correct statements from the following.

A. Detritivores like earthworms break down detritus into smaller particles by a process called fragmentation.
B. Water-soluble inorganic nutrients go down into the soil horizon and get precipitated as available salts by the process of leaching.
C. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances by catabolism.
D. Decomposition rate is slower if detritus is rich in lignin and chitin.
E. Warm and moist environments favour decomposition.

Choose the **correct** option.

(1) A, B, C, D only
(2) A, C, D, E only
(3) B, C, D only
(4) A, C, E only

112. In grasses, the guard cells enclosing the stomatal pore are:

(1) bean-shaped.
(2) spherical.
(3) barrel-shaped.
(4) dumb-bell shaped.

113. In the inheritance of which of the following traits, the genotypic and phenotypic ratio at F_2 of a Mendelian monohybrid cross was same?

(1) Flower colour in garden pea
(2) Seed colour in garden pea
(3) Flower colour in dogflower
(4) Skin colour in humans

114. Select the **incorrectly** matched pair.

(1) Phycomycetes - Sexual spore is zygospore
(2) Ascomycetes - Asexual spores are conidia
(3) Basidiomycetes - Sexual spores are basidiospores
(4) Deuteromycetes - Asexual spores are aplanospores

115. Arrange the following in ascending order of their size.

A. DNA content in *E. coli*.
B. Typical nucleosome.
C. One turn of DNA helix.
D. Diploid content of human DNA.
E. DNA of $\phi \times 174$ bacteriophage.

Choose the **correct** option.

(1) C → B → E → A → D
(2) D → C → B → E → A
(3) A → E → C → B → D
(4) D → B → C → A → E

116. *Cycas* does not have;

(1) tap roots.
(2) branched stems.
(3) coralloid roots.
(4) pinnate leaves.

117. Which of the following are outbreeding devices adapted by plants?

A. Pollen release and stigma receptivity are not synchronised.
B. Anther and stigma are placed at different positions.
C. Production of unisexual flowers.
D. Self-incompatibility.
E. Production of cleistogamous flowers.

Choose the **correct** option.

(1) A, B and D only
(2) A, C, D and E only
(3) A, B, C and D only
(4) C, D and E only

118. How many turns of Calvin cycle are required to form one molecule of glucose?
(1) 6 (2) 18
(3) 12 (4) 10

119. Identify the **incorrect** statement.
(1) Early man could easily perceive the difference between inanimate matter and living organisms.
(2) Early man deified some of the inanimate matter (wind, sea, fire etc.) and some among the animals and plants.
(3) All present day living organisms are related to each other and also to all organisms that ever lived on this earth.
(4) Societies which indulged in anthropocentric view of biology could register vast progress in biological knowledge.

120. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): A very significant contribution of mitosis is cell repair.
Reason (R): Mitotic divisions in the meristematic tissues result in a continuous growth of plants throughout their life.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

121. A structural feature of an ecosystem is;
(1) species composition.
(2) nutrient cycling.
(3) energy flow.
(4) productivity.

122. Match **List-I** with **List-II**.

List-I (Organism)		**List-II (Representation of sex chromosomes)**	
(A)	Male Grasshopper	(I)	XX
(B)	Female Grasshopper	(II)	ZW
(C)	Male Bird	(III)	XO
(D)	Female Bird	(IV)	ZZ

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-I, C-IV, D-III
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-II, C-III, D-IV

123. A farmer wants to induce flowering in mango, synchronise fruit-set in pineapples, and promote female flowers in cucumbers. Which single plant growth regulator would be most suitable for all these purposes?
(1) Auxin
(2) Gibberellin
(3) Cytokinin
(4) Ethylene

124. Introduction of Nile perch into Lake Victoria in east Africa led to the extinction of more than 200 species of;
(1) catfish *Clarias*.
(2) cichlid fish.
(3) Steller's sea cow.
(4) *Balanus*.

125. The stem is photosynthetic in;
(1) *Euphorbia* and *Opuntia*.
(2) cucumber and pumpkins.
(3) mint and jasmine.
(4) banana and *Chrysanthemum*.

126. The concept *'Omnis cellula-e cellula'*, which modified the cell theory, was first proposed by:
(1) Matthias Schleiden.
(2) Theodore Schwann.
(3) Rudolf Virchow.
(4) Robert Brown.

127. The number of autosomes is 45 in case of;
(1) a normal individual.
(2) Down's syndrome.
(3) Turner's syndrome.
(4) Klinefelter's syndrome.

[Contd...

128. Match **List-I** with **List-II**.

List-I (Scientists)		List-II (Experimental material)	
(A)	Frederick Griffith	(I)	*Streptococcus pneumoniae*
(B)	Meselson and Stahl	(II)	*Vicia faba*
(C)	Taylor and colleagues	(III)	*E. coli*
(D)	T. H Morgan	(IV)	*Drosophila melanogaster*

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-III, B-II, C-I, D-IV
(4) A-I, B-III, C-II, D-IV

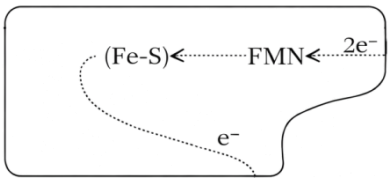
129. The flower is epigynous in the members of family;

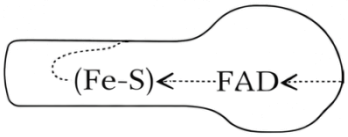
(1) Compositae.
(2) Poaceae.
(3) Cruciferae.
(4) Leguminosae.

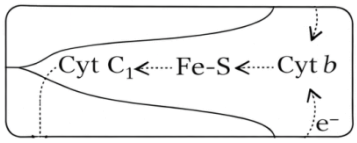
130. Under normal conditions, _____ are the most important factors influencing population density.

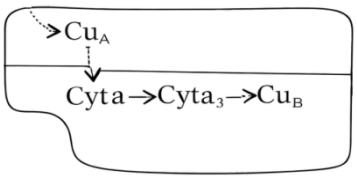
(1) natality and immigration
(2) mortality and emigration
(3) births and deaths
(4) immigration and emigration

131. In the mitochondrial ETS, complex II is represented as;

(1) 

(2) 

(3) 

(4) 

132. The cells of which of the following wall layers of a microsporangium generally have more than one nucleus?

(1) Epidermis
(2) Endothecium
(3) Middle layers
(4) Tapetum

133. The viroids;

A. lack genetic material.
B. lack capsid.
C. are smaller than viruses.
D. can pass through bacteria-proof filters.
E. have RNA of low molecular weight.

Choose the **correct** option.

(1) A, C, D only
(2) A, B, C, D only
(3) A, B, C, E only
(4) B, C, D, E only

134. Match **List-I** with **List-II**.

List-I (Structure)		List-II (Associated with)	
(A)	Pyrenoids	(I)	Pteridophytes
(B)	Prothallus	(II)	Bryophytes
(C)	Protonema	(III)	Green algae
(D)	Holdfast	(IV)	Brown algae

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-II, C-IV, D-I

135. Given below are two statements:

Statement I: Curd is rich in vitamin B_{12}.

Statement II: For the formation of curd, milk proteins are completely digested by acids produced by lactic acid bacteria.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

136. Which of the following set of animals have open circulatory system?
(1) Prawn, Cockroach and Scorpion
(2) Earthworm, Prawn and Octopus
(3) Frog, Lizard and Pigeon
(4) Shark, Rohu and Whale

137. Which of the following statements is/are **incorrect**?
A. Guanine attached to ribose sugar is called guanosine.
B. Thymidine contains ribose sugar.
C. Addition of a phosphate group to cytidine forms cytidylic acid.
D. DNA and RNA are polymers of nucleosides.
(1) A and C only (2) B and D only
(3) A, B and D only (4) B, C and D only

138. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Acrosome	(I)	Secretes Progesterone
(B)	Corpus Luteum	(II)	Enzymatic penetration of Ovum
(C)	Leydig Cells	(III)	Forms the placenta
(D)	Chorionic Villi	(IV)	Synthesis of Androgens

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-III, D-IV

139. In cockroach, flexibility of body movement between adjacent segments is primarily due to the presence of:
(1) Arthrodial membrane joining tergites and sternites.
(2) Chitinous thickening between two tergites.
(3) Fusion of dorsal and ventral sclerites.
(4) Muscular extensions of sternites.

140. Which of the following contraceptive methods function by altering the hormonal balance or releasing hormones into the body?
A. Vaults and Cervical Caps
B. LNG-20
C. Coitus interruptus
D. Multiload 375
E. Implants
(1) B, C and E only
(2) A, B and D only
(3) B and E only
(4) C, D and E only

141. If the Vital Capacity of a person is 4800 mL, Inspiratory Reserve Volume (IRV) is 2800 mL and Expiratory Reserve Volume (ERV) is 1200 mL, what is the Tidal Volume (TV)?
(1) 600 mL
(2) 700 mL
(3) 800 mL
(4) 900 mL

142. Which of the following statements regarding downstream processing is **correct**?
(1) It is identical for all biotechnological products.
(2) It does not require quality control testing.
(3) Products such as drugs must undergo clinical trials.
(4) It is completed before the formation of the final product.

143. During blood clotting, which of the following enzymes converts soluble fibrinogen into insoluble fibrin?
(1) Renin
(2) Epinephrine
(3) Thrombokinase
(4) Thrombin

144. Active reabsorption of sodium ions in the nephron predominantly occurs in which of the following regions?
(1) Bowman's capsule
(2) Descending limb of Henle's loop
(3) Distal convoluted tubule
(4) Proximal convoluted tubule

145. RNA interference (RNAi) technology is primarily used in biotechnology to:
(1) Develop pest-resistant plants against nematode infestation.
(2) Improve mineral absorption efficiency in plants.
(3) Reduce post-harvest losses in crops.
(4) Develop plants tolerant to abiotic stresses.

146. Which one of the following pairs of elements together constitute more than 80% of the human body weight?
(1) Oxygen and Carbon
(2) Oxygen and Hydrogen
(3) Carbon and Hydrogen
(4) Oxygen and Nitrogen

[Contd...

147. Choose **incorrect** statement?
(1) Juxtaglomerular cells of kidney secrete erythropoietin.
(2) Gastrin stimulates secretion of hydrochloric acid and pepsinogen.
(3) Cholecystokinin stimulates bile release from gall bladder.
(4) Growth factors are secreted only by endocrine glands.

148. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Thymus	(I)	Primary site of B-lymphocyte maturation
(B)	Peyer's patches	(II)	Mucosa-associated lymphoid tissue in respiratory, digestive and urogenital tract
(C)	Bone marrow	(III)	Differentiation and maturation of T-lymphocytes
(D)	MALT	(IV)	Secondary lymphoid nodules in ileum region of alimentary canal

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-I, C-II, D-III

149. Which of the following statement is **incorrect** regarding type of ovarian follicle shown in given below diagram?



(1) It contains a large fluid-filled antrum.
(2) The secondary oocyte is surrounded by zona pellucida and corona radiata.
(3) It secretes high levels of progesterone before ovulation.
(4) It bulges out on the surface of the ovary prior to ovulation and release secondary oocyte on ovulation.

150. Choose the option which contains only Australian marsupials.
(1) Sugar glider, Bandicoot, Wombat, Banded anteater
(2) Marsupial rat, Tiger cat, Lemur, Flying squirrel
(3) Flying phalanger, Marsupial mole, Bobcat, Wolf
(4) Lemur, anteater, Koala, Tiger cat

151. A reduction in cardiac output is most likely caused by which of the following?
(1) Parasympathetic stimulation
(2) Secretion of adrenaline
(3) Secretion of nor-adrenaline
(4) Sympathetic nerve activity

152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: People consuming tobacco are more likely to develop cardiovascular diseases.
Reason R: Nicotine stimulates the adrenal glands to release adrenaline and nor-adrenaline, leading to increased blood pressure and heart rate.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

153. The study of the history of life forms on Earth is known as:
(1) Paleontology
(2) Evolutionary Biology
(3) Systematics
(4) Physiology

154. The lungs in humans are surrounded by which of the following membranes?
(1) Periosteum
(2) Perichondrium
(3) Pericardium
(4) Pleural membrane

155. The restriction endonuclease *Hind*III was isolated from *Haemophilus influenzae* strain Rd.
Based on the standard nomenclature, which of the following statements is **correct**?
(1) 'Hin' represents the strain and 'd' represents the genus.
(2) 'H' represents genus, 'in' represents species, 'd' represents strain, and 'III' indicates order of isolation.
(3) 'Hind' represents species and 'III' represents strain.
(4) 'Hi' represents genus and species, while 'nd' represents strain.

[Contd...

156. For the successful production of thousands of plants through micropropagation, the plant cells must possess which specific intrinsic property?

(1) Biofortification
(2) Totipotency
(3) Differentiation
(4) Somatic Hybridisation

157. Consider the following statements:

Statement I: Zygote Intra Fallopian Transfer (ZIFT) is a procedure where the zygote is transferred into the uterus.

Statement II: Intra Uterine Transfer (IUT) involves transfer of embryos with more than 8 blastomeres in to uterus.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

158. Match **List-I** with **List-II**.

List-I (Common name)		**List-II (Scientific name)**	
(A)	Sea-pen	(I)	*Physalia*
(B)	Sea fan	(II)	*Gorgonia*
(C)	Brain coral	(III)	*Meandrina*
(D)	Portuguese Man-of-war	(IV)	*Pennatula*

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-IV, C-II, D-I

159. Exophthalmic goitre is also known as:

(1) Acomegaly
(2) Graves' disease
(3) Cretinism
(4) Addison's disease

160. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Micro-injection	(I)	Bombardment of plant cells with high velocity particles coated with DNA
(B)	Biolistics	(II)	Used as carriers coated with DNA for plant transformation
(C)	Disarmed pathogen vector	(III)	Direct delivery of recombinant DNA into nucleus of animal cell
(D)	Gold or tungsten micro-particles	(IV)	Transfers recombinant DNA by infecting the host cell

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

161. Given below are two statements:

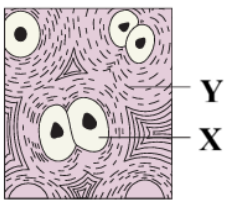
Statement I: Passive immunity provides immediate relief as it involves injection of preformed antibodies.

Statement II: Tight junctions are found in epithelial tissues and they provide cementing to the neighbouring cells.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

162. In the given diagram of specialized connective tissue correctly identify the labelled part **X** and **Y.**



(1) **X** – Collagen fibres, **Y**- Chondrocyte
(2) **X** – Collagen fibres, **Y**- Osteocyte
(3) **X** –Osteocyte, **Y**- Collagen fibres
(4) **X** – Chondrocyte, **Y**- Collagen fibres

[Contd...

163. Match **List-I** with **List-I** w.r.t. *Plasmodium*.

List-I		List-II	
(A)	Sporozoite	(I)	Female *Anopheles* mosquito
(B)	Gametocyte	(II)	Infective stage to humans transmitted by female *Anopheles* mosquito
(C)	Primary host	(III)	Sexual stage taken up by mosquito during blood meal
(D)	Secondary host	(IV)	Human

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-IV, D-III

164. Which of the following is **not** a method of spread of venereal diseases?
(1) Unprotected sexual contact with an infected person.
(2) Transfusion of contaminated blood.
(3) Sharing infected needles.
(4) Sharing clothes and utensils with an infected person.

165. Read the following statements (A - E) regarding the treatment of Adenosine Deaminase (ADA) deficiency via Gene therapy and select the **correct** ones.
A. The first clinical gene therapy was given in 1990 to a 4-year-old girl with ADA deficiency.
B. This disorder is caused due to the deletion of the gene for the enzyme adenosine deaminase.
C. In the procedure, lymphocytes from the patient's blood are grown in culture and a functional ADA cDNA is introduced into them using a pBR322 vector.
D. Enzyme replacement therapy and bone marrow transplantation are considered completely curative methods for all patients.
E. A permanent cure is possible if the gene isolated from marrow cells producing ADA is introduced into cells at early embryonic stages.
(1) A, B, C and E only
(2) A, C and D only
(3) A, B and E only
(4) A, B, C, D and E

166. Which of the following options **correctly** represents the characteristic features of phylum Coelenterata?
(1) Triploblastic, segmented body and bilaterally symmetrical
(2) Triploblastic, flattened body and acoelomate condition
(3) Diploblastic, mostly marine and radially symmetrical
(4) Triploblastic, unsegmented body and bilaterally symmetrical

167. Which of the following hormone is **not** involved in the regulation of spermatogenesis?
(1) FSH
(2) LH
(3) Androgens
(4) Aldosterone

168. An alien DNA fragment introduced into a bacterial cell fails to multiply even though it is not degraded. The most likely reason is that the DNA fragment:
(1) Lacks recognition sites for restriction enzymes.
(2) Is inserted into the cytoplasm instead of the nucleus.
(3) Is present in multiple copies inside the host cell.
(4) Is not linked to a DNA sequence capable of initiating replication.

169. Which of the following animals is **not** viviparous?
(1) *Ornithorhynchus*
(2) *Macropus*
(3) *Pteropus*
(4) *Elephas*

170. With reference to oogenesis, consider the following statements:
A. Oogenesis is initiated during embryonic development.
B. Oogonia continue to be formed even after birth.
C. Primary oocytes are arrested at prophase-I of meiosis.
D. Each primary oocyte is surrounded by granulosa cells to form a primary follicle.
E. At puberty, only about 60,000–80,000 primary follicles remain in both ovaries.
Which of the above statements are **correct**?
(1) A, C, D and E only
(2) A, B, C and D only
(3) B and E only
(4) A, C and D only

171. The genetic modification in *Bt* Cotton specifically involves the introduction of which of the following genes to provide resistance against Cotton Bollworms?
(1) *cry*IAb and *cry*IIAb
(2) *cry*IAc and *cry*IIAb
(3) *cry*IAb only
(4) *cry*IIAc and *cry*IAb

172. Which of the following statements **correctly** describes the quaternary structure of a protein?
(1) It refers to the sequence of amino acids in a polypeptide chain.
(2) It refers to the folding of a single polypeptide into α-helix or β-sheet.
(3) It refers to the spatial arrangement of multiple folded polypeptide subunits.
(4) It refers to the interaction between amino acids via peptide bonds only.

173. Which of the following cancer treatments have side effects like hair loss, anemia etc.?
(1) Surgery
(2) Radiation therapy
(3) Chemotherapy
(4) Immunotherapy

174. In frog, cutaneous respiration is possible because the skin is:
(1) Thick and keratinized.
(2) Dry and glandless.
(3) Moist and richly supplied with blood capillaries.
(4) Covered with scales.

175. The observation that populations remain stable in size despite their capacity to grow exponentially suggests that:
(1) There is no variation among individuals.
(2) There is intense competition for limited resources.
(3) Reproduction in nature is always minimal.
(4) Acquired characters are inherited.

176. The site where a motor neuron transmits an impulse to a skeletal muscle fibre is known as:
(1) Sarcoplasmic reticulum
(2) Neuromuscular junction
(3) Transverse tubule
(4) Myofibril

177. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Apoenzyme	(I)	Non-protein part tightly bound to enzyme
(B)	Competitive inhibitor	(II)	Protein part of enzyme
(C)	Coenzyme	(III)	Organic compound transiently bound to apoenzyme
(D)	Prosthetic group	(IV)	Closely resembles the substrate in its molecular structure

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-II, B-IV, C-III, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-II, C-I, D-III

178. Excessive urine formation (diuresis) is inhibited by the action of which of the following hormones?
(1) Renin secreted by JG cells through inactivation of osmoreceptors
(2) Atrial natriuretic factor (ANF) released from the atria
(3) Aldosterone secreted from the adrenal medulla
(4) Vasopressin released from the neurohypophysis

179. In frog, which of the following statements is **correct** regarding the entry of digestive secretions into the duodenum?
(1) Bile and pancreatic juice enter separately into the stomach.
(2) Bile enters the duodenum directly from liver.
(3) Pancreatic juice enters the rectum before bile.
(4) Bile and pancreatic juice enter the duodenum through a common bile duct.

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The resting axonal membrane is more permeable to K^+ ions than Na^+ ions.
Reason R: The sodium-potassium pump transports 3 Na^+ inwards and 2 K^+ outwards.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

■■■

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